

4. Properties of Convolution

- **Algebraic properties of convolution**
 - (1) **Commutative Property**
 - (2) **Distributive Property and Parallel Systems**
 - (3) **Associative Property and Series Systems**
- **Time-Shift Property**
- **Convolution with Impulse/Step Functions**
- **Differentiation & Integration Properties**

4. Continuous-Time Convolution

4.1 Algebraic properties of convolution

(1) Commutative property

$$f_1(t) * f_2(t) = f_2(t) * f_1(t)$$

proof:

$$f_1(t) * f_2(t) = \int_{-\infty}^{\infty} f_1(\tau) \cdot f_2(t - \tau) d\tau$$

let $t - \tau = \lambda$

Thus $\tau = t - \lambda$

$$d\tau = -d\lambda$$

$$\tau : \int_{-\infty}^{\infty} \rightarrow \lambda : \int_{\infty}^{-\infty}$$

$$= \int_{\infty}^{-\infty} f_2(\lambda) \cdot f_1(t - \lambda) (-d\lambda)$$

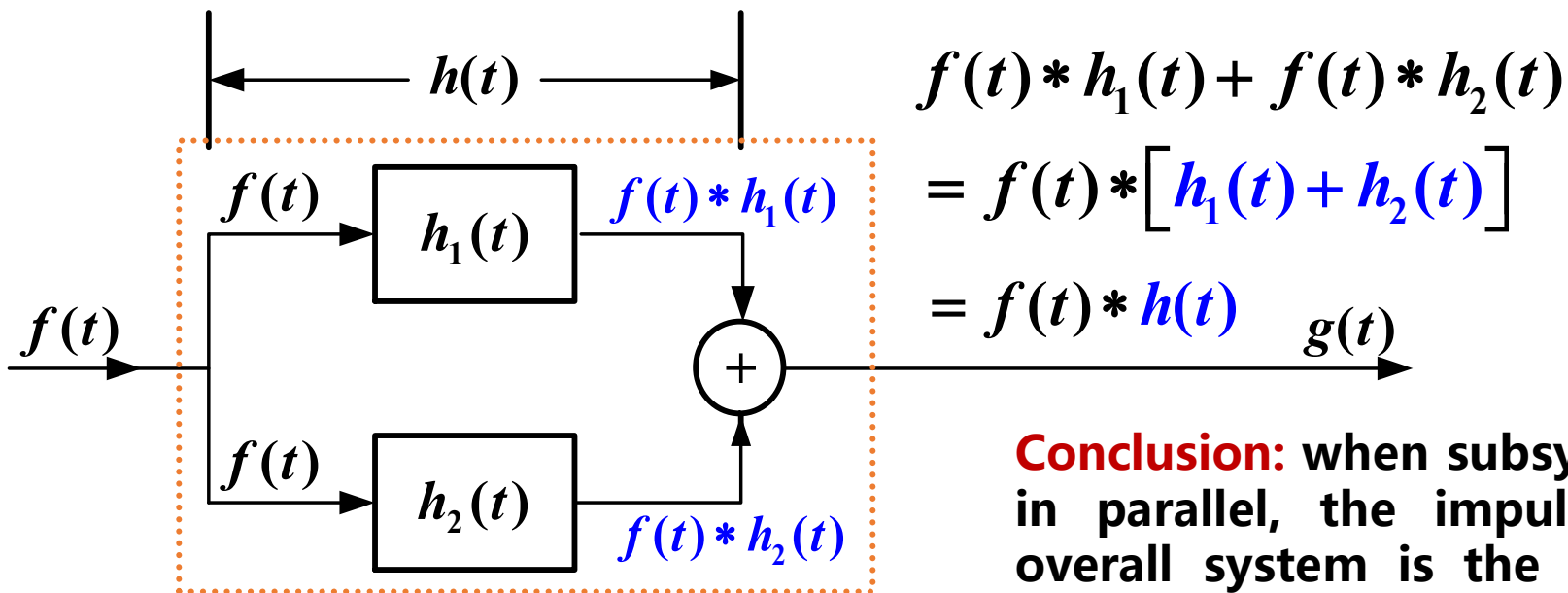
$$= \int_{-\infty}^{\infty} f_2(\lambda) \cdot f_1(t - \lambda) d\lambda$$

$$= f_2(t) * f_1(t)$$

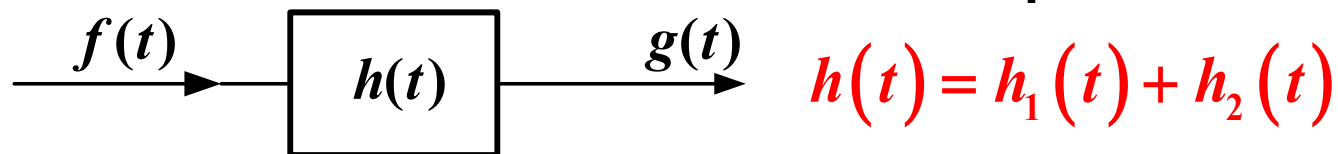
4.1 Algebraic properties of convolution

(2) Distributive Property and Parallel Systems

$$f_1(t) * [f_2(t) + f_3(t)] = f_1(t) * f_2(t) + f_1(t) * f_3(t)$$



Conclusion: when subsystems are connected in parallel, the impulse response of the overall system is the sum of the impulse responses of the individual subsystems

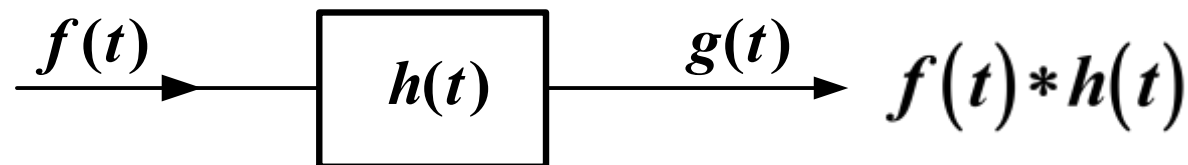
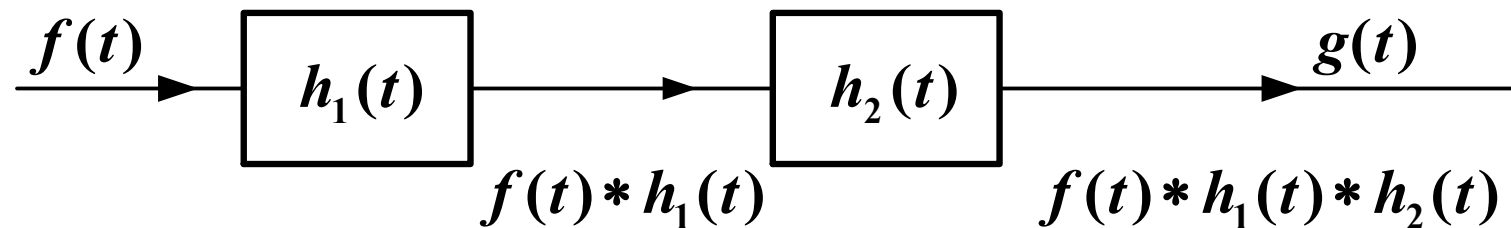


4.1 Algebraic properties of convolution

(3) Associative Property and Series Systems

$$f(t) * h_1(t) * h_2(t) = f(t) * [h_1(t) * h_2(t)]$$

The system is connected in series, and the block diagram shows:



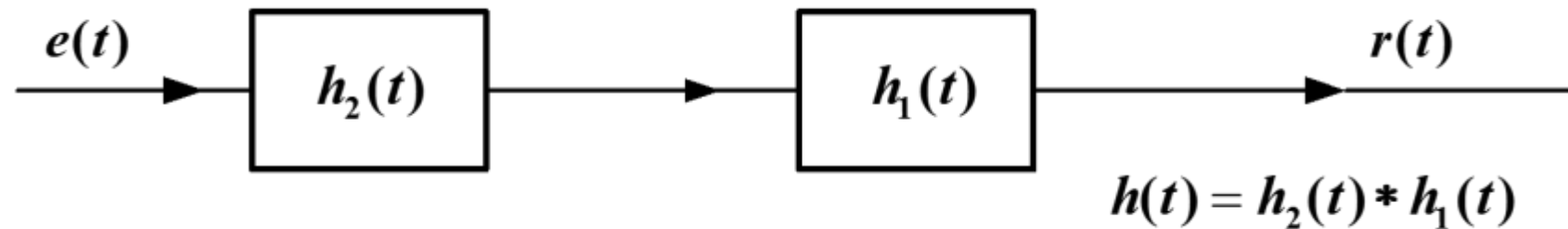
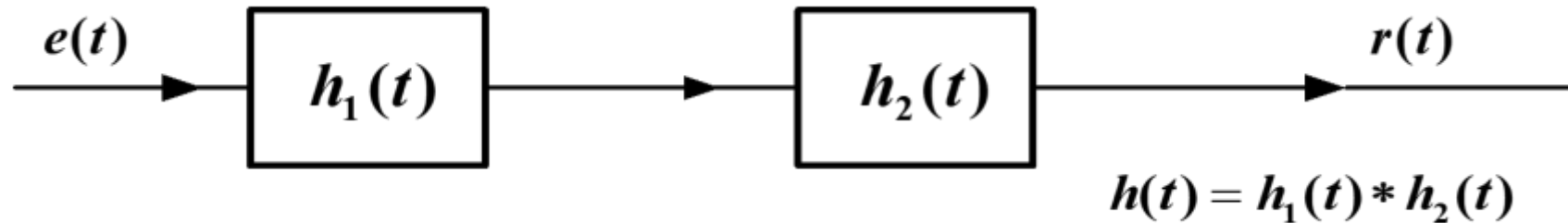
$$h(t) = h_1(t) * h_2(t)$$

Conclusion: when subsystems are connected in series (cascade), the impulse response of the overall system is the convolution of the individual impulse responses

4. Continuous-Time Convolution

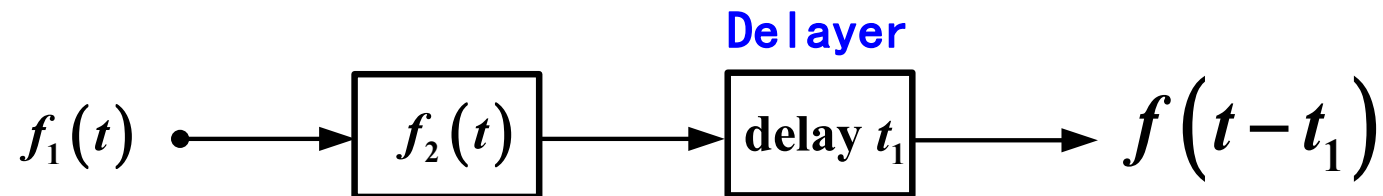
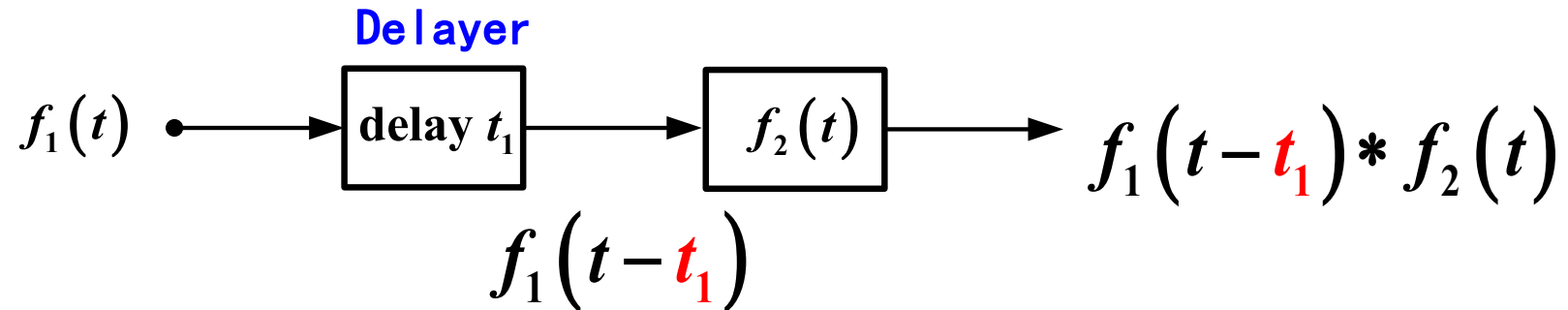
4.1 Algebraic properties of convolution

The order of subsystems connected in series (cascade) can be swapped



4.2 Time-Shift Property

If $f(t) = f_1(t) * f_2(t)$, Then $f_1(t - \mathbf{t}_1) * f_2(t) = f(t - \mathbf{t}_1)$



$$f(t) = f_1(t) * f_2(t)$$

$$f_1(t - \mathbf{t}_1) * f_2(t - \mathbf{t}_2) = f(t - \mathbf{t}_1 - \mathbf{t}_2)$$

4.3 Convolution with Impulse/Step Functions

a. $f(t) * \delta(t) = \int_{-\infty}^{\infty} f(t - \tau) \delta(\tau) d\tau$

$$= f(t) \int_{-\infty}^{\infty} \delta(\tau) d\tau = f(t)$$

$$f(t) \xrightarrow[h(t) = \delta(t)]{1} f(t)$$

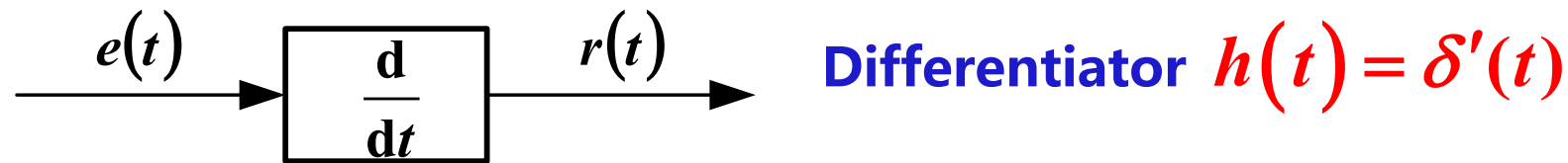
The unit Impulse response of an ideal conductor is $\delta(t)$

b. $f(t) * \delta(t - t_0) = f(t - t_0)$



4.3 Convolution with Impulse/Step Functions

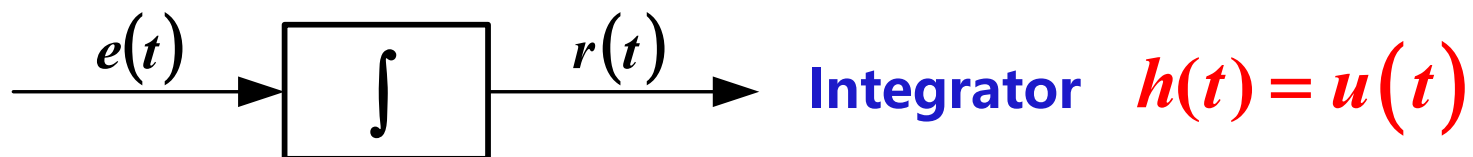
c. $f(t) * \delta'(t) = f'(t)$



$f(t) * \delta^{(k)}(t) = f^{(k)}(t)$ **K numbers differentiators**

$f(t) * \delta^{(k)}(t - t_0) = f^{(k)}(t - t_0)$ **Differential and Time Shift**

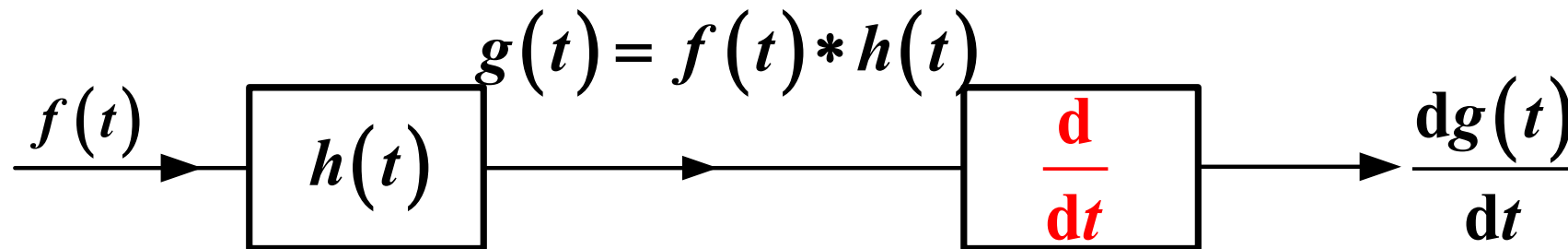
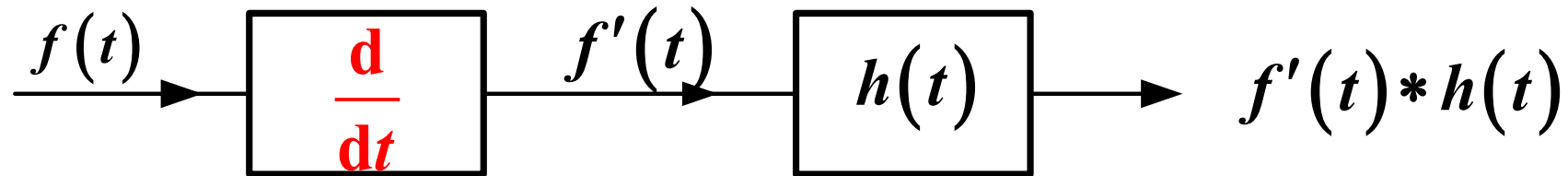
d. $f(t) * u(t) = \int_{-\infty}^t f(\lambda) d\lambda$



4.4 Differentiation & Integration Properties

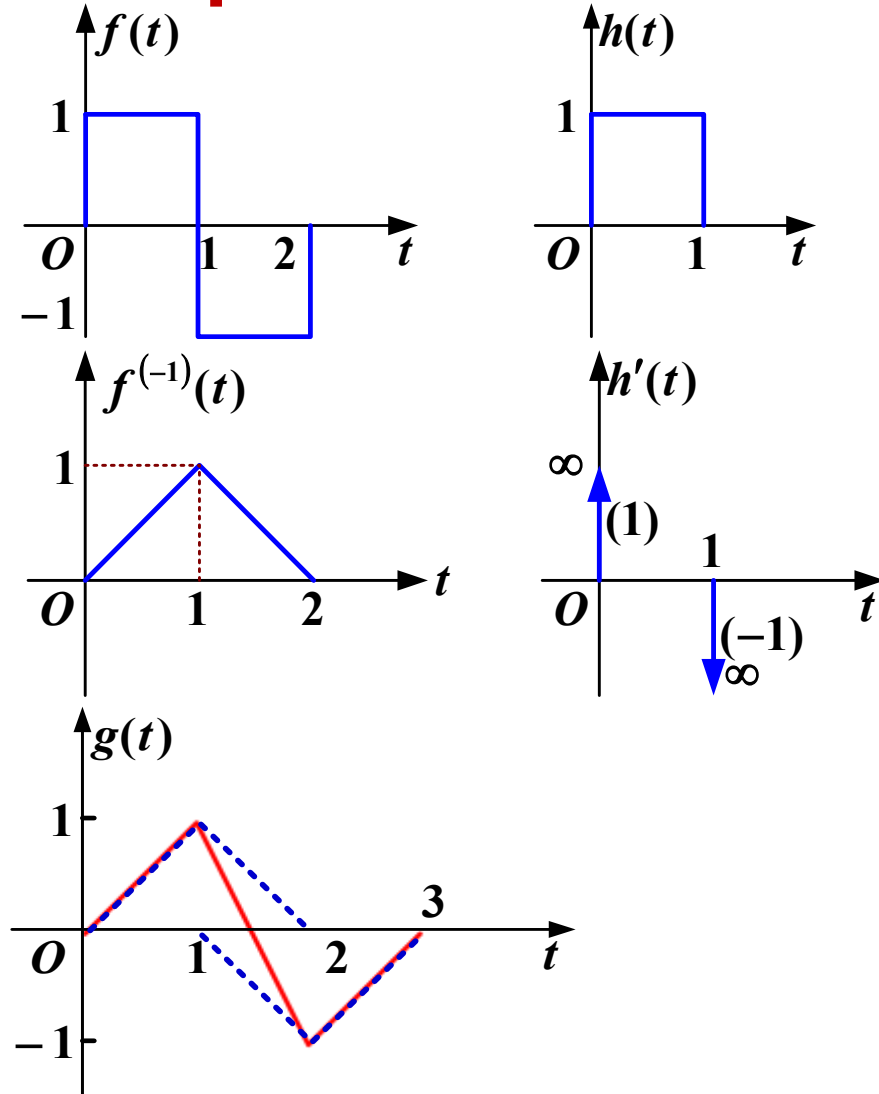
If $g(t) = f(t) * h(t)$

Then $g'(t) = f'(t) * h(t) = f(t) * h'(t)$



4. Continuous-Time Convolution

Example 1



If $f(t), h(t)$, get $g(t) = f(t) * h(t)$

$$g(t) = f^{(-1)}(t) * h'(t)$$

$$= f^{(-1)}(t) * [\delta(t) - \delta(t-1)]$$

$$= f^{(-1)}(t) - f^{(-1)}(t-1)$$

Convolution with $\delta(t)$: a weighted and shifted version of the original signal

4. Continuous-Time Convolution

5. Convolution Calculation-Analytical Method

$$r_{zs}(t) = \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau$$

Due to the causality of the system or the time-limited nature of the excitation signal, the limits of integration in convolution may vary. Determining the **correct integration limits** in convolution integrals is crucial

4. Continuous-Time Convolution

Example 2

Given $e(t) = u(t) - u(t - t_0)$,

$h(t) = e^{-t} u(t)$,

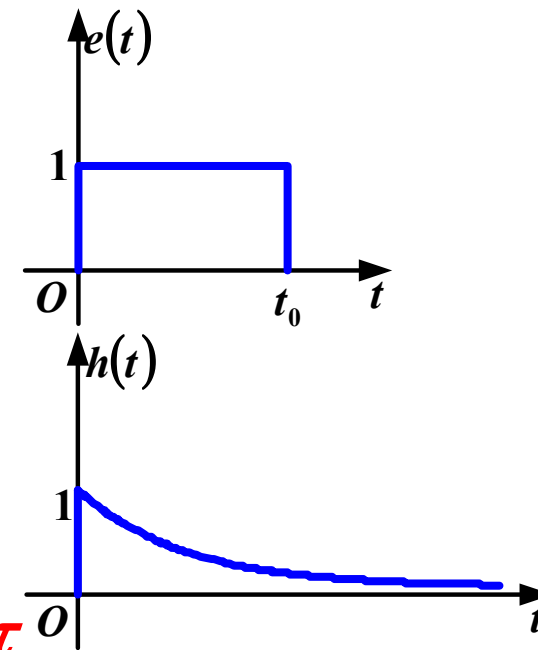
Solve $r(t) = e(t) * h(t)$, Let $t_0 > 0$.

$$r(t) = \int_{-\infty}^{\infty} e(\tau) \cdot h(t - \tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau) - u(\tau - t_0)] \cdot e^{-(t-\tau)} u(t - \tau) d\tau$$

$$= e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau) u(t - \tau)] d\tau - e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau - t_0) u(t - \tau)] d\tau$$

$$= r_1(t) - r_2(t)$$



4. Continuous-Time Convolution

Example 2 Determining Integration Limits (Crucial)

$$r_1(t) = e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau)u(t-\tau)] d\tau$$

$$= \left[e^{-t} \int_0^t e^{\tau} d\tau \right] \cdot u(t)$$

$$= (1 - e^{-t}) u(t)$$

$$u(\tau) \cdot u(t - \tau) \neq 0$$

$$\Rightarrow \begin{cases} \tau > 0 \\ t - \tau > 0 \end{cases}$$

$$\Rightarrow \begin{cases} 0 < \tau < t \\ t > 0 \end{cases}$$

4. Continuous-Time Convolution

Example 2

$$r_2(t) = e^{-t} \int_{-\infty}^{\infty} e^{\tau} [u(\tau - t_0)u(t - \tau)] d\tau$$

$$= \left[e^{-t} \int_{t_0}^t e^{\tau} d\tau \right] \cdot u(t - t_0)$$

$$= [1 - e^{-(t-t_0)}] u(t - t_0)$$

$$u(\tau - t_0) \cdot u(t - \tau) \neq 0$$

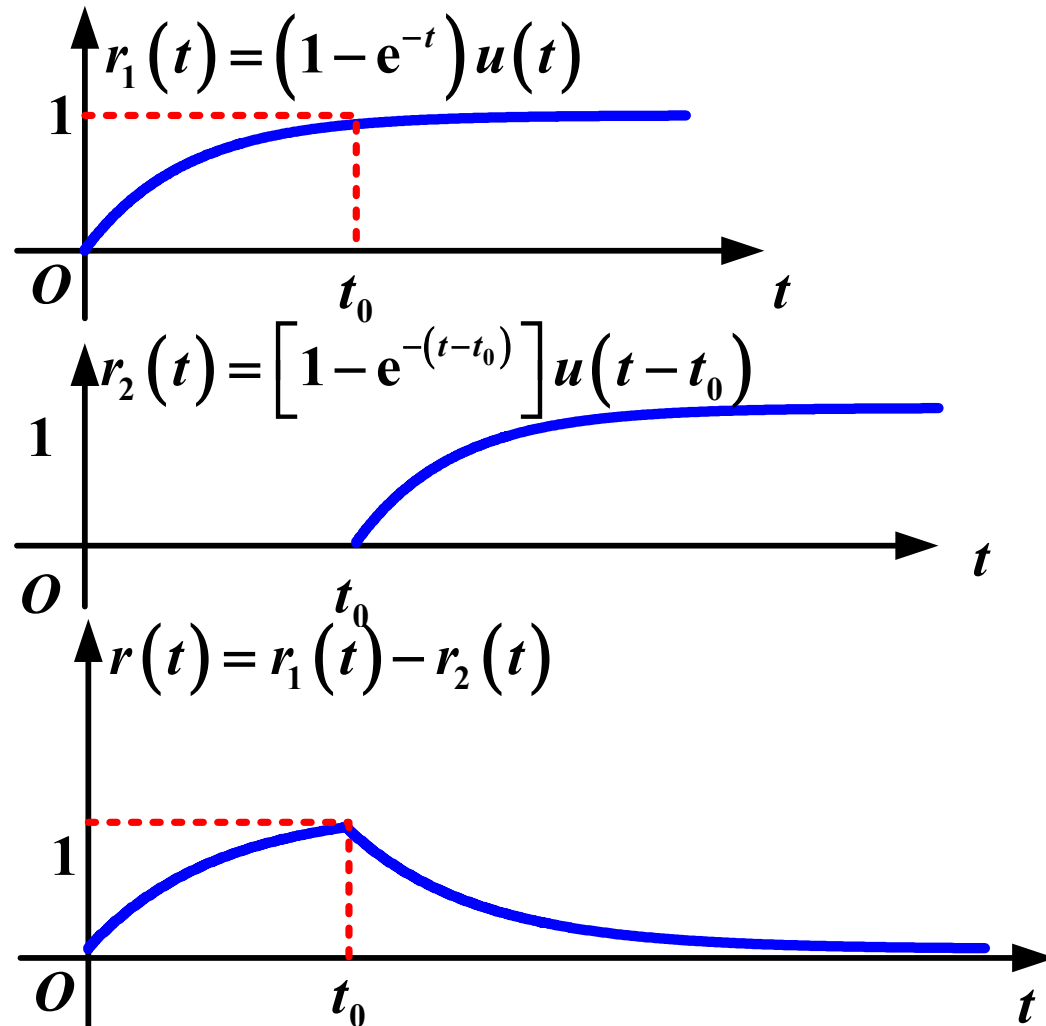
$$\Rightarrow \begin{cases} \tau - t_0 > 0 \\ t - \tau > 0 \end{cases}$$

$$\Rightarrow \begin{cases} t_0 < \tau < t \\ t > t_0 \end{cases}$$

4. Continuous-Time Convolution

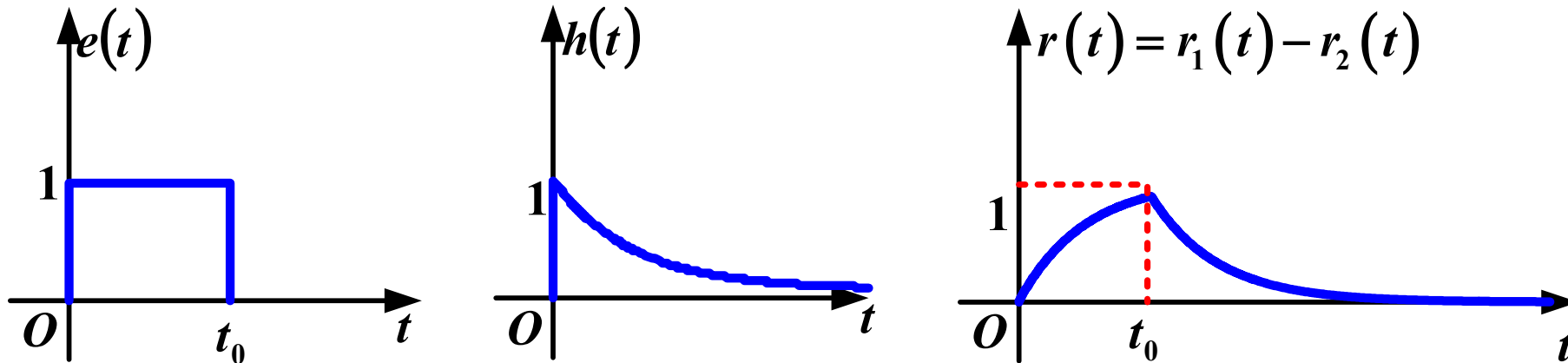
Example 2

$$\begin{aligned} r(t) &= r_1(t) - r_2(t) \\ &= (1 - e^{-t})u(t) \\ &\quad - [1 - e^{-(t-t_0)}]u(t-t_0) \end{aligned}$$



4. Continuous-Time Convolution

Example 2



- The output exhibits a tendency to "catch up" with the input
- This is a system with memory, where the output depends not only on the current input but also on past inputs

5. Convolution Calculation-Graphical Method

When the functional formula is complex, using the **graphical method** makes it easier to intuitively determine the **integration limits**

$$g(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

1. $f_1(t) \rightarrow f_1(\tau)$, **change the integral variable to τ**

2. $f_2(t) \rightarrow f_2(\tau) \xrightarrow{\text{reversal}} f_2(-\tau) \xrightarrow{\text{time delay}} f_2(t - \tau)$

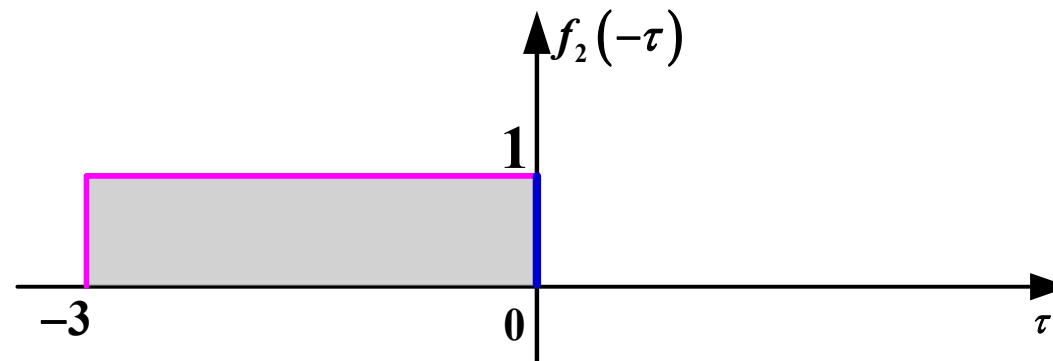
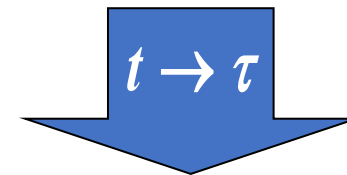
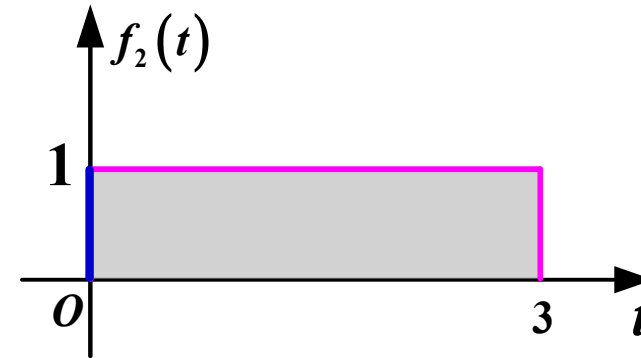
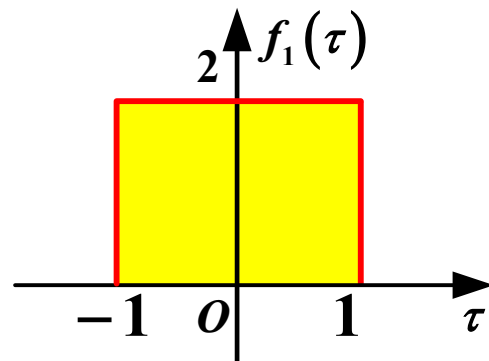
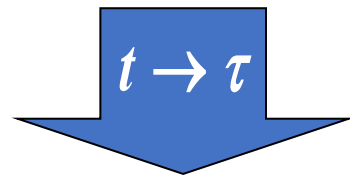
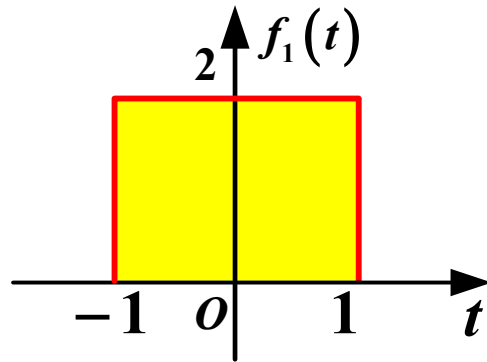
3. **Multiplication** $f_1(\tau) \cdot f_2(t - \tau)$

4. **Integral of product** $\int_{-\infty}^{\infty} f_1(\tau) \cdot f_2(t - \tau) d\tau$

4. Continuous-Time Convolution

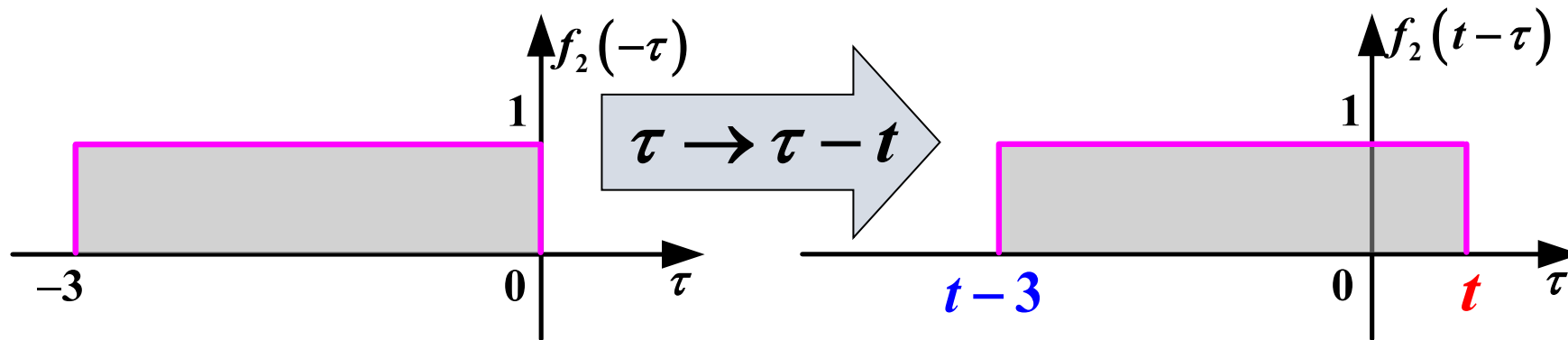
Example 3

The waveforms of $f_1(t)$ and $f_2(t)$ are as follows, find the convolution of $f_1(t)$ and $f_2(t)$, $g(t)$



4. Continuous-Time Convolution

Example 3



$$f_2(t-\tau) = f_2[-(\tau-t)]$$

Original coordinate system New coordinate system

$$\tau = 0$$

$$\tau = t$$

$$\tau = -3$$

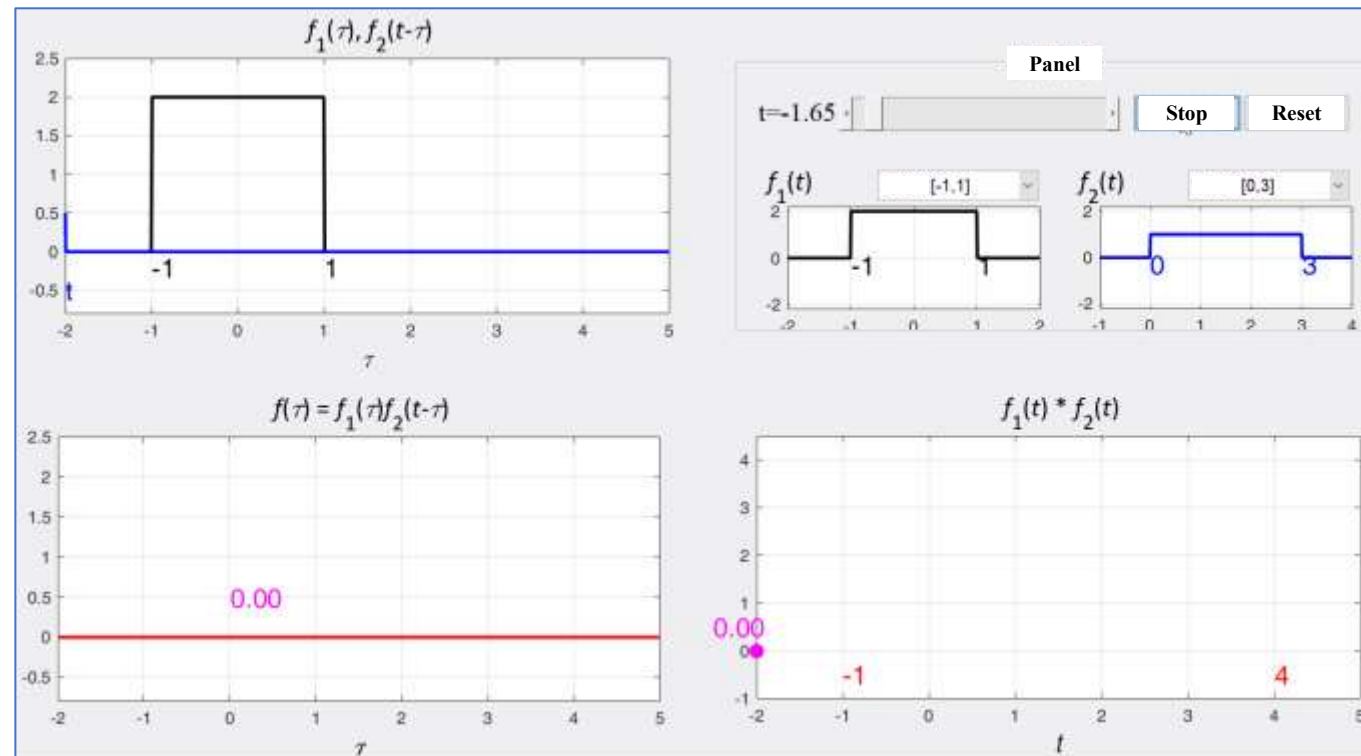
$$\tau = t-3$$

4. Continuous-Time Convolution

Example 3

$$g(t) = \int_{-\infty}^{\infty} f_1(\tau) f_2(t - \tau) d\tau$$

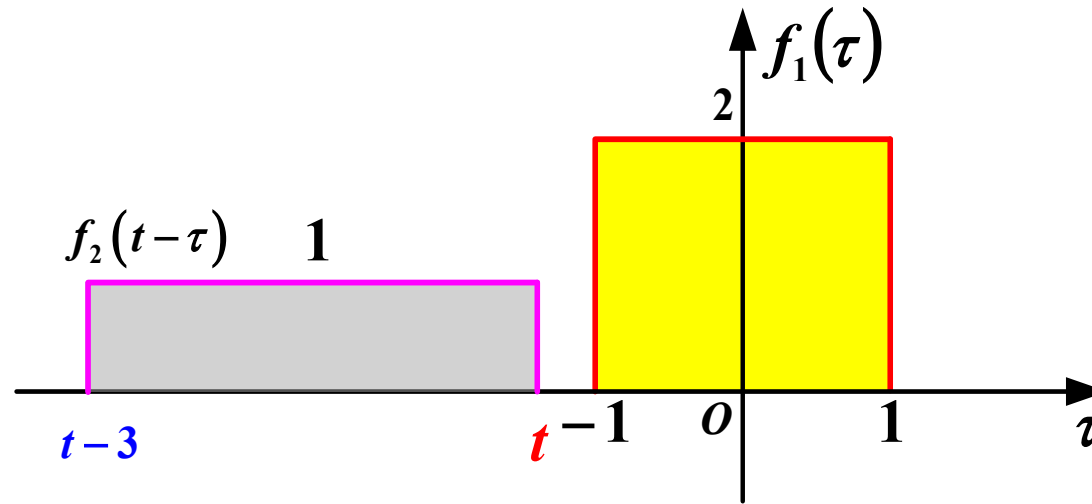
t from $-\infty$ to $+\infty$ corresponds to $f(t-\tau)$ move from left to right



**Multiply and
then integrate**

4. Continuous-Time Convolution

Example 3



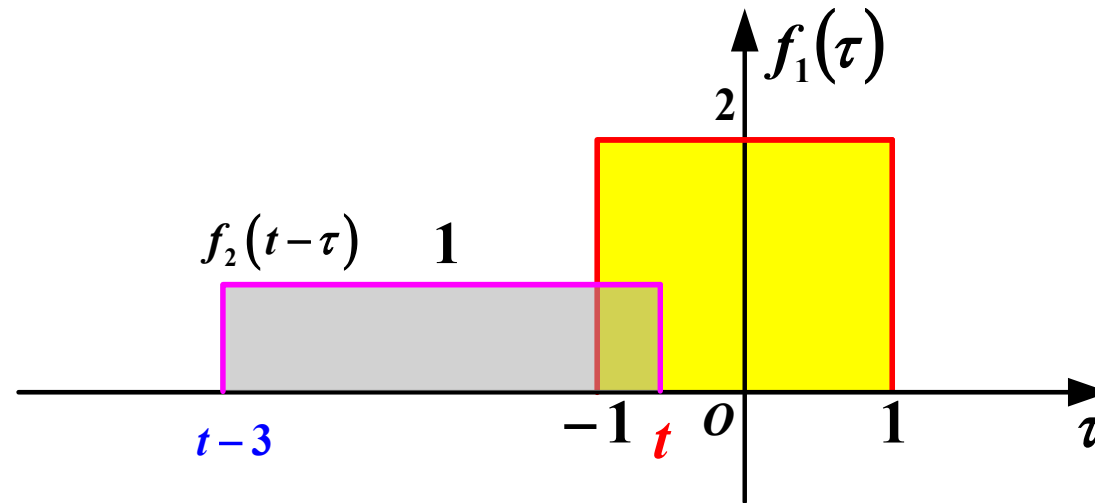
① $t \leq -1$

$$f_1(\tau) \cdot f_2(t-\tau) = 0$$

$$g(t) = f_1(t) * f_2(t) = 0$$

4. Continuous-Time Convolution

Example 3

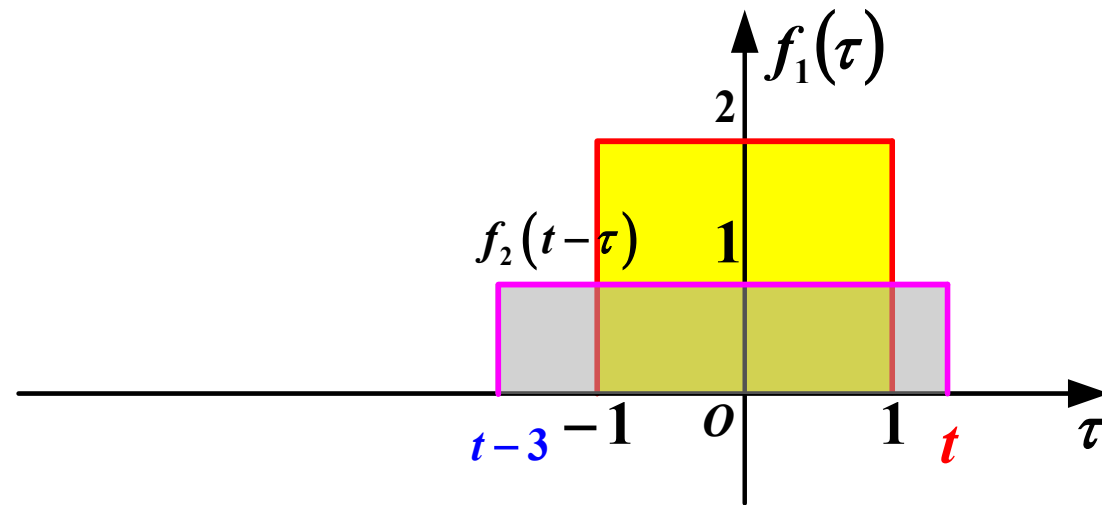


$$\textcircled{2} \quad \begin{cases} t \geq -1 \\ t \leq 1 \end{cases} \Rightarrow -1 \leq t \leq 1$$

$$\begin{aligned} g(t) &= \int_{-1}^t f_1(\tau) \cdot f_2(t-\tau) d\tau \\ &= \int_{-1}^t 2 d\tau = 2t \Big|_{-1}^t = 2t + 2 \end{aligned}$$

4. Continuous-Time Convolution

Example 3

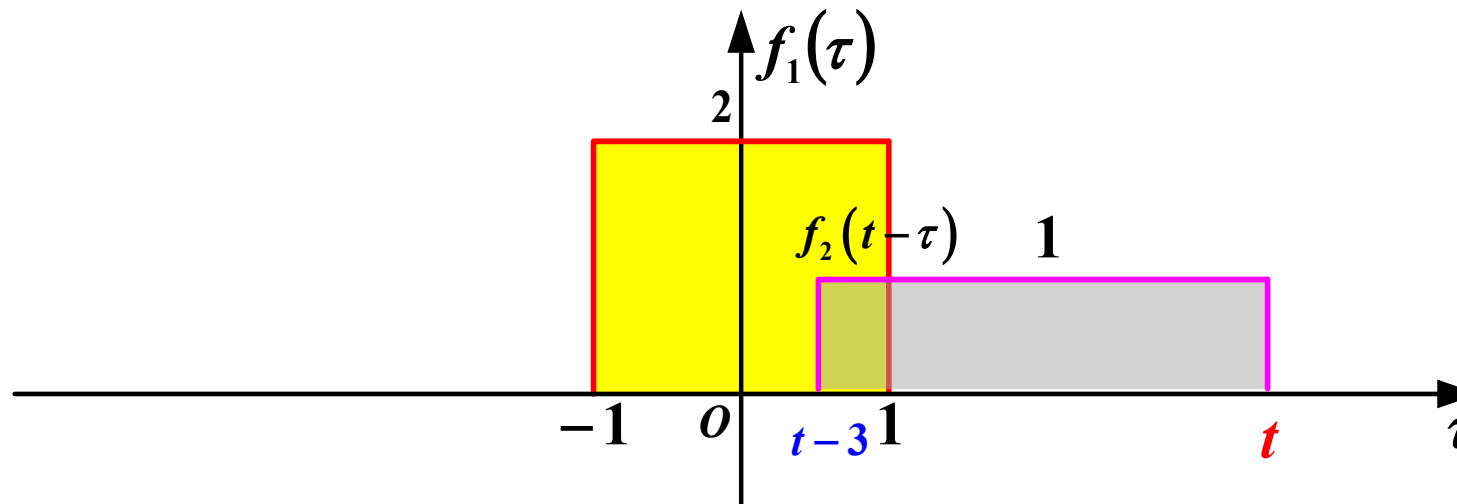


$$\textcircled{3} \begin{cases} t \geq 1 \\ t-3 \leq -1 \end{cases} \text{ Thus } 1 \leq t \leq 2$$

$$\begin{aligned} g(t) &= \int_{-1}^1 2 d\tau \\ &= 4 \end{aligned}$$

4. Continuous-Time Convolution

Example 3

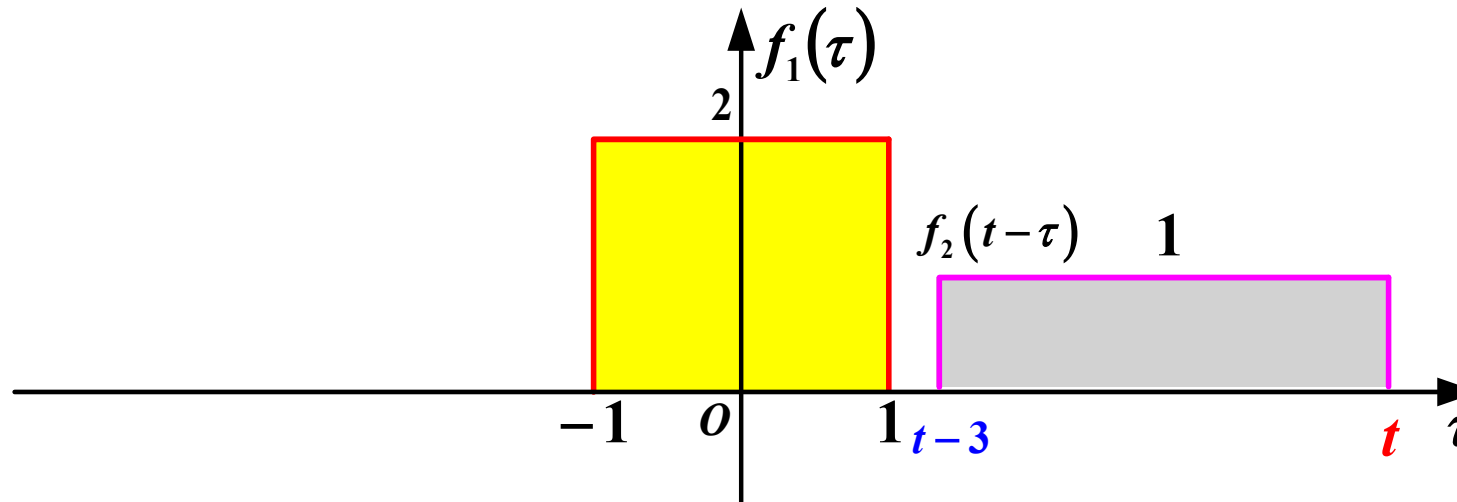


$$\textcircled{4} \begin{cases} t-3 \geq -1 \\ t-3 \leq 1 \end{cases} \quad \text{Thus } 2 \leq t \leq 4$$

$$\begin{aligned} g(t) &= \int_{t-3}^1 2 \, d\tau = 2\tau \Big|_{t-3}^1 \\ &= -2t + 8 \end{aligned}$$

4. Continuous-Time Convolution

Example 3

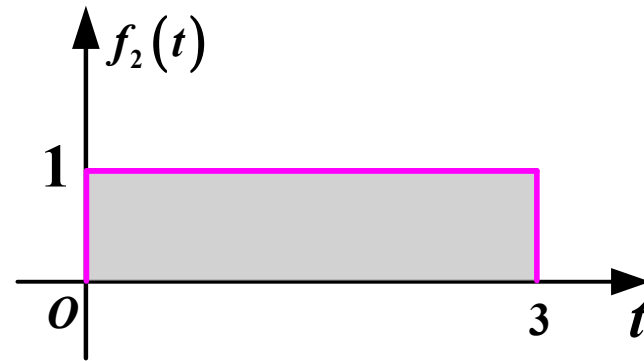
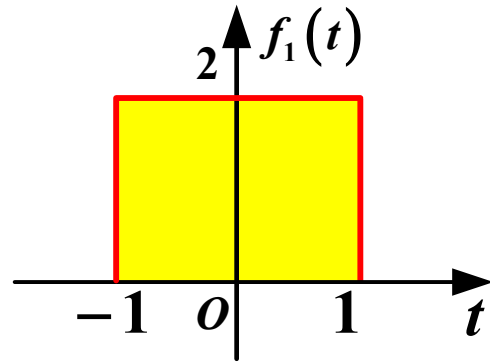


⑤ $t - 3 \geq 1$ Thus $t \geq 4$

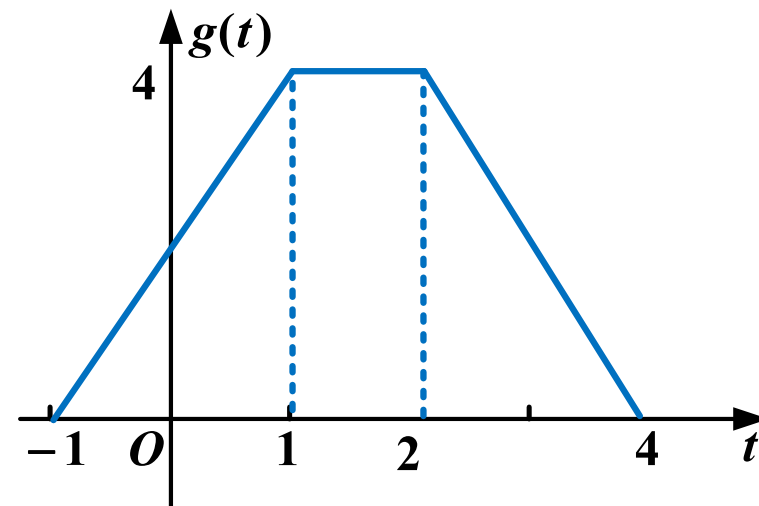
$$g(t) = 0$$

4. Continuous-Time Convolution

Example 3

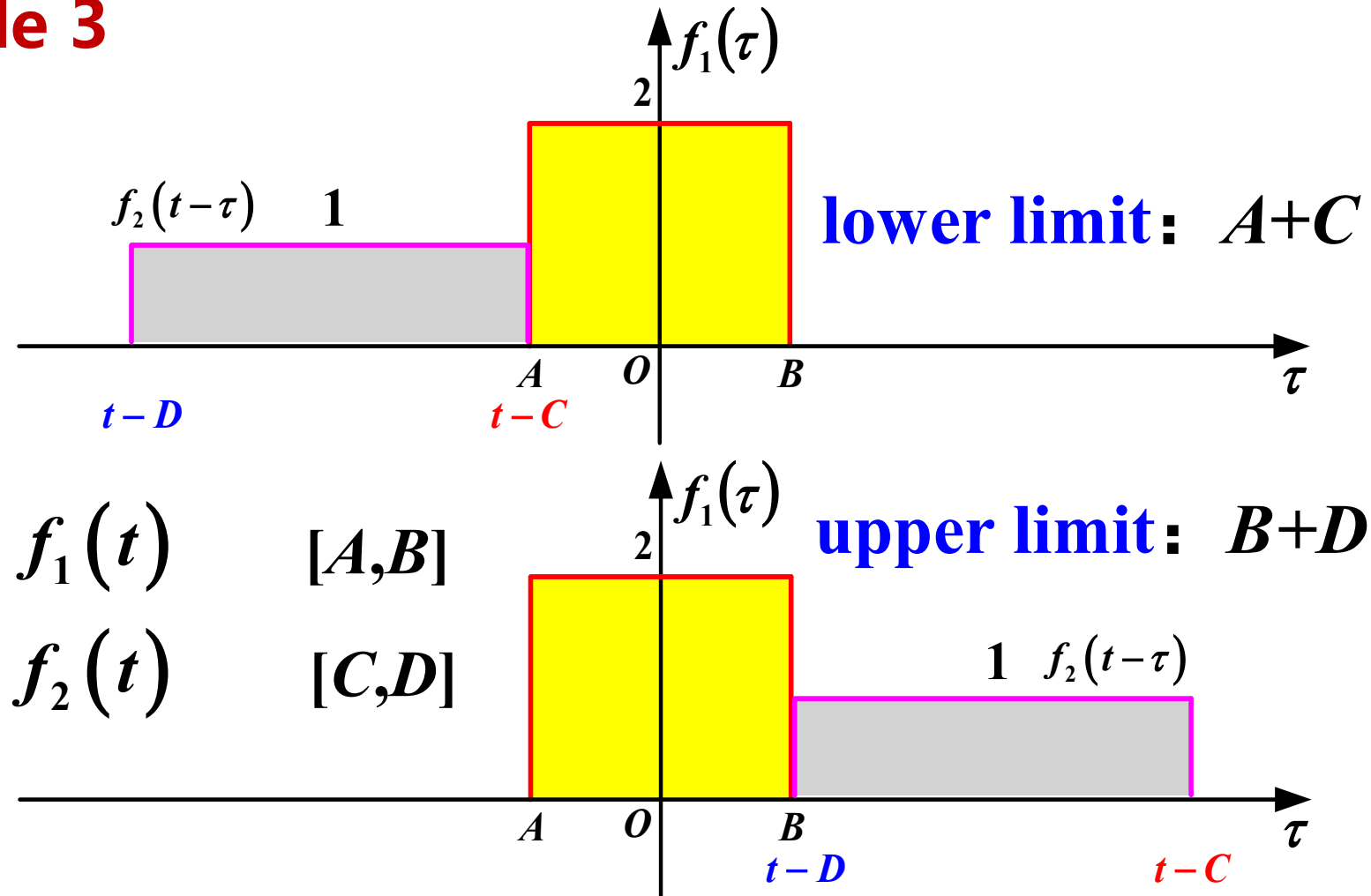


$$g(t) = \begin{cases} \textcircled{2} = 2t + 2 & -1 \leq t \leq 1 \\ \textcircled{3} = 4 & 1 \leq t \leq 2 \\ \textcircled{4} = -2t + 8 & 2 \leq t \leq 4 \\ \textcircled{1} \textcircled{5} = 0 & \text{else } t \end{cases}$$



4. Continuous-Time Convolution

Example 3



The width of $g(t)$ = the sum of the widths of the two signals

Summary

1. Introduction to Convolution
2. Zero-State Response via Convolution
3. Physical Interpretation of Convolution
4. Properties of Convolution
5. Convolution Calculation

$$\begin{aligned}r(t) &= e(t) \otimes h(t) \\&= e(t) * h(t) \\&= \int_{-\infty}^{\infty} e(\tau) h(t - \tau) d\tau\end{aligned}$$